

Ian Leung

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EDUCATION

University of Waterloo

Expected 2028

Bachelor of Computer Science, Finance, & Statistics – Triple Major

GPA: 3.8/4.0

- President's Scholarship of Distinction (\$5000), Undergraduate Achievement Award (\$1500)
- Relevant Coursework: Object-Oriented Programming (Advanced), Sequential Programming (Advanced), Functional Programming (Advanced), Interpreters (Advanced), Linear Algebra, Calculus, Probability, Statistics, Proofs, Financial Markets & Data Analytics

WORK EXPERIENCE

Roblox

Jun 2026 – Present

Software Engineer Intern (Network Reliability) | Go, Python

San Mateo, CA

- Automated the Network Engineer push process from scratch, pushing configs to **5,000+** devices in global datacenters and reducing the configuration drift window by **99%**
- Orchestrated GitHub Actions (CI/CD) and Temporal workflows for safe, autonomous deployment of configuration changes, using canary and gradual rollout strategies to save **30+ engineering hours** per month
- Integrated **LLM analysis** of commit diffs and per-device health checks to ensure staged changes are safe to deploy, preventing outages/misconfigurations and identifying issues on **15+** network devices
- Identified and fixed a resource leak in an internal network service, preventing API lockout and recovering **7%** CPU utilization across devices

Waterloo Quant Club

Jan 2026 – Present

Core Developer (DevOps) | Rust, Python, AWS

Waterloo, ON

- Engineered portal.waterlooquantclub.com, a full-stack portal hosted on AWS for **700+** members, supporting JWT authentication, custom application forms, resume uploads, reviewer workflows, and Excel exports
- Solo-developed a Grafana observability platform using PostgreSQL, TLS, and AWS, providing real-time monitoring of **100+** metrics and automatic **Discord bot alerting** for critical service failures
- Built and deployed [2v2bluff](#), a real-time multiplayer **Rust/WebSocket** card game supporting four concurrent players

Sunnybrook Research Institute

Jul 2025 – Aug 2025

Software Research Intern | Python

Toronto, ON

- Built an event tracing and playback tool to simulate medical professionals' interactions with **Focused Ultrasound** software, enabling realistic testing and diagnosis of software defects, unintended user behaviors, and edge cases
- Created an automated software testing pipeline that generates synthetic and replayable JSON event flows, reducing software testing times by **91%**
- Developed a logging service to validate test results against baselines, catching **4** software defects before production release

SKILLS

Languages Python, Java, C++, C#, C, Go, Kotlin, TypeScript, SQL, HTML, CSS, Bash

Technologies Temporal, gRPC, REST, PyTorch, Pandas, Grafana, PostgreSQL, MongoDB, React, Actions

Tools Linux, Git, Docker, AWS (EC2, S3, Lambda, etc.), OpenRouter, Jupyter, Supabase, Android Studio

PROJECTS

PeerAssist – Peer Review Platform | Java, Supabase, Apache PDFBox, Maven

- Created a full-stack **Java** desktop application for **20+** students to upload, review, and grade peers' academic work
- Integrated a **Supabase** backend for authentication and data storage, enforcing **Row-Level Security** policies across profiles, documents, and reviews tables to control per-user access
- Rendered uploaded PDFs with **Apache PDFBox** and added search, sorting, and ranking by average review score; packaged the app into a single zero-config executable JAR via **Maven** shade plugin

Trust No Ghost – 3D Maze Exploration Game | Unity, C#, Itch.io

- Shipped **Trust No Ghost** for **60+ users** at the UW Game Jam Fall 2025, placing in the **top 10** most rated games
- Harnessed Unity and C# to develop expanding maze designs, AI pathfinding algorithms, and dynamic moving mechanics